

ONE-PAGE FORMULA SHEET

CHAPTER 4 · Consumer Theory

Budget: $p_1 \cdot q_1 + p_2 \cdot q_2 = I$ | Slope = $-p_1/p_2$

Optimum: $MRS = MU_1/MU_2 = p_1/p_2$

Cobb-Douglas: $q_1^* = [a/(a+b)] \cdot I/p_1$

Labor-Leisure: $c = wL + V$; $MU_L/MU_c = w$

Intertemporal: $c_1 + c_2/(1+r) = I_1 + I_2/(1+r)$

IT optimum: $MU_1/MU_2 = 1+r$

Price elasticity: $E_p = (P/Q)(dQ/dP)$

Income elasticity: $E_I = (I/Q)(dQ/dI)$

Cross-price: $E_{xy} = (P_y/Q_x)(dQ_x/dP_y)$

CHAPTER 9 · Uncertainty & Risk

$E[x] = \sum(\pi_i \cdot x_i)$

$E[u] = \sum(\pi_i \cdot u(x_i))$

Risk averse: $u'' < 0$ | neutral: $u''=0$ | loving: $u''>0$

Certainty equivalent: $u(CE) = E[u]$

Risk premium: $RP = E[x] - CE$

Fair bet: $E[\text{payoff}] = 0$

Fair insurance: $P_{\text{fair}} = E[\text{loss}]$

Max premium: $E[u \text{ with ins}] = E[u \text{ no ins}]$

Insurance RP: $P_{\text{max}} - P_{\text{fair}}$

Diversify: negative correlation lowers variance